

Course Code: B23EE4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ELECTRIC VEHICLES					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define gradeability of a vehicle.	1	2	2
	b).	Define feasibility of EVs.	1	3	2
	c).	What are the main powertrain components of an EV?	2	2	2
	d).	List the merits of Hybrid Electric Vehicles?	2	3	2
	e).	Define State of Charge (SOC)?	3	3	2
	f).	Illustrate the importance of battery recycling.	3	3	2
	g).	Define electric drive system?	4	3	2
	h).	What is electronic commutation in BLDC motors?	4	3	2
	i).	Differentiate between static and dynamic charging.	5	3	2
	j).	List few EV charging standards.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain in detail evolution of EV's?	1	2	5
	b).	Differentiate EVs with IC engine vehicle.	1	3	5
		OR			
3.	a).	Interpret the impact of design considerations on the performance of electric vehicles.	1	4	5
	b).	Analyze the effect of different vehicle load forces on the performance of an electric vehicle.	1	4	5
		UNIT-2			
4.	a).	Illustrate various configurations of Hybrid EVs in detail?	2	3	5
	b).	Assess EV motor sizing in detail.	2	4	5
		OR			
5.	a).	Illustrate in detail about power train components of EV?	2	3	5
	b).	Assess the working of Fuel cell Electric Vehicle and give its merits over BEV?	2	4	5

		UNIT-3			
6.	a).	Analyze different types of battery technologies in energy storage based system of Electric Vehicles?	3	3	5
	b).	Justify the presence and performance of Battery Management System in Electric Vehicles?	3	4	5
		OR			
7.	a).	Examine the construction and working of a Li-ion battery with a neat diagram?	3	4	5
	b).	What is Fuel Cell? Explain the operation of fuel cell? Write advantages and disadvantages of fuel cells?	3	4	5
		UNIT-4			
8.	a).	Explain the Electric Drive components in detail?	4	3	5
	b).	Explain the working of Permanent Magnet BLDC motor drive?	4	4	5
		OR			
9.	a).	Explain the operation of a Switched Reluctance Motor (SRM) in an Electric Vehicle?	4	3	5
	b).	Compare PMSM and BLDC motor drive?	4	4	5
		UNIT-5			
10.	a).	Explain the basic requirements for charging system in EV?	5	3	5
	b).	Illustrate the Architecture of EV Charger with schematic	5	3	5
		OR			
11.	a).	Explain in detail V2G and V2V Technology?	5	3	5
	b).	Illustrate the effect of EV load on the Grid?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23HS4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ENERGY MANAGEMENT AND AUDIT					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List out various types of energy available	1	2	2
	b).	What is primary energy? Give one example.	1	2	2
	c).	Mention any two qualities of a successful energy manager.	2	2	2
	d).	Name two principles of energy management.	2	2	2
	e).	Define power factor.	3	2	2
	f).	Define Life Cycle Cost (LCC).	3	2	2
	g).	List the benefits of DSM.	4	2	2
	h).	What is load priority technique?	4	2	2
	i).	Define cost index in energy management.	5	2	2
	j).	What is the purpose of a pie chart in energy auditing?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Write a detailed note on renewable and non-renewable sources of energy.	1	3	5
	b).	Review the energy scenario in India brief	1	3	5
		OR			
3.	a).	Explain the Indian scenario for renewable energy.	1	3	5
	b).	Describe the concept of energy planning.	1	3	5
		UNIT-2			
4.	a).	Elaborate the functions and qualities required in an energy manager.	2	3	5
	b).	Explain the role of energy audit in an energy management programme.	2	3	5
		OR			
5.	a).	Describe the necessary steps involved in implementing an effective energy management programme.	2	3	5
	b).	Discuss the principles of energy management.	2	3	5

		UNIT-3			
6.	a).	Discuss Equivalent Annual Worth (EAW) and how it aids in comparing investment alternatives	3	3	5
	b).	What are energy-efficient motors? Explain types, selection criteria, and economic impact.	3	3	5
		OR			
7.	a).	Discuss the causes and corrective methods of low power factor. Also, explain benefits of PF improvement.	3	3	5
	b).	Describe methods for achieving energy-efficient lighting in commercial and industrial buildings.	3	3	5
		UNIT-4			
8.	a).	Describe different techniques of DSM.	4	3	5
	b).	Discuss Time of day models for planning.	4	3	5
		OR			
9.	a).	Discuss load management techniques used in DSM	4	3	5
	b).	Explain peak clipping and peak shifting.	4	3	5
		UNIT-5			
10.	a).	Explain the objectives and importance of energy audits in industries.	5	3	5
	b).	Describe the use and significance of Sankey diagrams in energy auditing.	5	3	5
		OR			
11.	a).	What are energy conservation schemes? Describe with examples from industrial or commercial sectors.	5	3	5
	b).	Explain the different types of energy audits and highlight their differences.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EE4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BATTERY MANAGEMENT & CHARGING SYSTEMS					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define a Battery Management System (BMS).	1	2	2
	b).	Name the major components of a microprocessor-based BMS.	1	2	2
	c).	Define State of Charge (SoC) of a battery.	2	1	2
	d).	What is meant by C-rate of a battery?	2	1	2
	e).	What is the need for battery protection in a BMS?	3	2	2
	f).	List various topological structures of a BMS	3	2	2
	g).	Define Level 2 EV charging.	4	1	2
	h).	Define constant current (CC) charging method.	4	1	2
	i).	What is a battery swapping station?	5	1	2
	j).	Compare domestic charging infrastructure and public charging infrastructure.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the need for a Battery Management System (BMS) in modern battery-powered applications.	1	3	5
	b).	Explain the operation of a microprocessor-based Battery Management System.	1	3	5
OR					
3.	a).	Illustrate the various stages involved in the development of a power battery management system with a neat sketch.	1	3	5
	b).	Explain the system failures that may occur due to the absence of a Battery Management System (BMS).	1	3	5
UNIT-2					
4.	a).	Explain the methods used for measurement of battery voltage, current, and temperature.	2	3	5
	b).	Explain the Equivalent Circuit Model (ECM) used for battery modelling.	2	3	5
OR					
5.	a).	Illustrate a battery equivalent circuit and explain the function of each component.	2	3	5
	b).	Describe the Coulomb Counting Method used for estimation of State of Charge (SoC).	2	3	5

		UNIT-3			
6.	a).	Explain balancing control methods used in Battery Management Systems.	3	3	5
	b).	Explain the master-slave topology used in Battery Management System with neat sketch.	3	3	5
		OR			
7.	a).	Illustrate the various communication interfaces used in a Battery Management System (BMS).	3	3	5
	b).	Explain the role of thermal management in improving battery performance and safety.	3	3	5
		UNIT-4			
8.	a).	Briefly explain the different charging technologies used in electric vehicles.	4	3	5
	b).	Explain the charging circuit of an On-Board Charger used in Level 1 charging.	4	3	5
		OR			
9.	a).	Illustrate the trickle charging method used in EV batterie with neat sketch.	4	3	5
	b).	Compare different power level types used for EV charging.	4	3	5
		UNIT-5			
10.	a).	Explain the features of public charging infrastructure used for EVs.	5	3	5
	b).	Illustrate the working of a battery swapping station in EV applications with neat sketch	5	3	5
		OR			
11.	a).	Explain the construction and operation of a fast-charging station.	5	3	5
	b).	Illustrate the working of a battery swapping station in EV applications with neat sketch	5	3	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EE4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
POWER SYSTEM OPERATION & CONTROL					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the coordination equation for thermal power plants	1	2	2
	b).	What is heat rate curve and cost curve? and draw with a neat sketch	1	2	2
	c).	Define Unit Commitment? and name the different techniques to solve U.C Problem.	2	2	2
	d).	What are the constraints in hydrothermal scheduling? and state the power balance equation in hydrothermal scheduling.	2	2	2
	e).	Draw block diagram of Single area system with neat sketch and represents the parameters	3	2	2
	f).	What is free governor operation and Write expression for steady state frequency?	3	2	2
	g).	Define Area Control Error (ACE).	4	2	2
	h).	Draw the two-area system with block diagram with tie-line modeling.	4	2	2
	i).	Define long term frequency dynamics and coherent area dynamics	5	2	2
	j).	What is Centre of Inertia (COI)? and write the equation for COI angle.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Derive general transmission line loss formula and state assumptions made in calculating B- coefficients.	1	3	5
	b).	A constant 30MW supplied by two 150MW generators, 1 and 2 for which the respective incremental fuel costs are $dC_1/dP_1=0.1P_1+20$, $dC_2/dP_2=0.2P_2+25$. With P in MW and C in Rs/h. Calculate (i) the most economical division of load between the generators (ii) the saving in Rs/day there by obtain compared to equal load sharing between the machines.	1	3	5
		OR			
3.	a).	Write step by step procedure for obtaining optimal scheduling of generating units by neglecting the transmission losses	1	3	5

	b).	A power system consists of two 200MW units whose input cost data are represented by the equations: $C_1 = 0.03P_1^2 + 21P_1 + 750$ Rs/hour, $C_2 = 0.5P_2^2 + 18P_2 + 980$ Rs/hour. If the total received power $P_R = 350$ MW, compute the load division between the units for the most economic operation.	1	3	5
		UNIT-2			
4.	a).	Describe the hydrothermal economic load scheduling. Derive the necessary equations?	2	3	6
	b).	Discuss about different constraints considered in solving a unit commitment problem	2	3	4
		OR			
5.	a).	With the help of flow chart, explain the solution of unit commitment problem using dynamic programming.	2	3	4
	b).	Using dynamic programming method to determine the most economical units to be committed to supply a load of 6 MW. There are four units with the following data $C_1=0.8 P_1^2 + 21 P_1$, $C_2=0.6P_2^2 + 22P_2$, $C_3=0.5P_3^2 + 21P_3$ and $C_4=0.6P_4^2 + 20P_4$. The maximum and minimum limits for each unit are 6 MW and 1 MW respectively.	2	3	6
		UNIT-3			
6.	a).	Explain the mathematical modeling of speed governing system and derive the transfer function of speed governor model. State the assumptions made	3	3	5
	b).	A 200 MVA synchronous generator is operating at 3000 rpm, 50Hz. A load of 40MW is suddenly applied to the machine and the steam valve of the turbine opens only after 0.4 sec due to the time lag in the generator action. Calculate the frequency to which the generated voltage drops before the steam flow commences to increase to meet the new load. Given that the value of H of the generator is 5.5 kW-sec/KVA of the generator capacity.	3	3	5
		OR			
7.	a).	Draw the block diagram representation of a single area system and derive the expression for static and dynamic response of system under uncontrolled case.	3	3	5
	b).	Two generators are operating in parallel with 4% and 5% drop characteristics of their respective governors sharing a load of 600MW. Compute the load shared by the machines and the system for this load. The normal frequency of the system was 50Hz.	3	3	5
		UNIT-4			
8.	a).	Explain the Automatic Voltage regulator with a neat sketch	4	3	5
	b).	Discuss the importance of combined load frequency control and economic dispatch control with a neat block diagram	4	3	5

		OR			
9.	a).	Obtain the mathematical modeling of tie line power in an interconnected system and its block diagram	4	3	5
	b).	Derive the steady state frequency deviation expression for two area LFC for a step load change in area-1? (assumption can take as per requirement)	4	3	5
		UNIT-5			
10.	a).	Explain the four operating states of Power system with a neat sketch	5	3	5
	b).	Explain briefly about the Stability Enhancement methods	5	3	5
		OR			
11.	a).	Explain the concepts of coherent area dynamic with a detailed example.	5	3	5
	b).	Explain preventive and emergency control in power systems.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUTONOMOUS VEHICLE SYSTEMS					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is a 2D rotation matrix and how is it used in autonomous vehicles?	1	2	2
	b).	Mention two key differences between control and decision-making modules in AVs.	1	2	2
	c).	What is the dynamic bicycle model	2	2	2
	d).	What are the states of an AV in the context of motion control?	2	2	2
	e).	What is brake-by-wire in autonomous vehicles?	3	2	2
	f).	Explain the working principle of LiDAR	3	2	2
	g).	Draw a simple FSM for basic driving behaviors	4	2	2
	h).	What is the difference between behavior planning and motion planning in autonomous vehicles?	4	2	2
	i).	What is a URDF file in ROS and why is it used?	5	2	2
	j).	Name any two commonly used real-time topics in ROS2 and their purpose.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the five levels of vehicle autonomy as defined by SAE. Highlight the differences in human involvement at each level.	1	3	5
	b).	Describe coordinate transformation between reference frames used in AVs	1	3	5
OR					
3.	a).	Explain the basic architecture of an autonomous vehicle. Label the components under sensors, perception, planning, control, and decision-making modules.	1	3	5
	b).	How does forward kinematics affect vehicle path tracking, and what constraints are imposed by the vehicle's kinematic model?	1	4	5

		UNIT-2			
4.	a).	Describe path tracking and list the main control objectives involved in motion control of autonomous vehicles.	2	3	5
	b).	Illustrate PID control for lane-following in autonomous vehicles and analyze its feedback mechanism using a block-diagram.	2	4	5
		OR			
5.	a).	Explain the difference between longitudinal and lateral vehicle dynamics.	2	3	5
	b).	What are ride comfort metrics in autonomous vehicles, and how do they influence motion control strategies?	2	4	5
		UNIT-3			
6.	a).	Compare LiDAR, Radar, and Camera sensors in autonomous vehicles based on their range, image clarity, and ability to work in different weather conditions.	3	3	5
	b).	Describe the role of Kalman Filter in state estimation. Use a block diagram to explain its two main steps: prediction and update.	3	4	5
		OR			
7.	a).	Illustrate a basic sensor fusion setup using LiDAR and camera, and explain how combining their data improves object detection.	3	3	5
	b).	Describe the architecture of a SLAM system used in autonomous navigation.	3	4	5
		UNIT-4			
8.	a).	Explain the use of Finite State Machines (FSMs) in autonomous driving. Design a simple FSM to model different driving behaviors such as cruising, stopping, and turning at intersections.	4	3	5
	b).	Discuss the role of decision trees in maneuver selection in autonomous vehicles. Create a sample decision tree for choosing between lane change, maintain lane, or slow down actions based on traffic conditions.	4	4	5
		OR			
9.	a).	Describe the Markov Decision Process (MDP) framework used in planning for autonomous vehicles. Illustrate with an example how states, actions, transition probabilities, and rewards define a driving scenario.	4	4	5
	b).	Explain reward engineering in reinforcement learning for autonomous vehicles, emphasizing the role of safety, comfort, and efficiency metrics in shaping the reward function.	4	4	5

		UNIT-5			
10.	a).	Explain the architecture of ROS2 and describe how nodes communicate using topics, services, and actions. Include a simple diagram.	5	3	5
	b).	Illustrate and explain the flow of data between sensors (LiDAR, camera, GPS) and ROS2 nodes. Use topics like /lidar/points, /camera/info, and /gps/fix in your explanation.	5	4	5
		OR			
11.	a).	Discuss the organization of ROS2 workspace and package management. Provide a step-by-step process for creating a custom package.	5	3	5
	b).	Describe the role of ROS_control and ROS2 driver interfaces in implementing a drive-by-wire system. Explain with a real-world example.	5	4	5
CO-COURSE OUTCOME		KL-KNOWLEDGE LEVEL	M-MARKS		

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE4106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
POWER ELECTRONIC DRIVES					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Define Electric Drive.	1	2	2
	b).	What are electric starting methods.	1	2	2
	c).	List the classification of load torques.	2	2	2
	d).	Calculate the moment of inertia of flywheel of mass 45kg and radius of 2m.	2	3	2
	e).	Write the expression for average output voltage of a single-phase full converter.	3	3	2
	f).	Why does a series motor have high starting torque?	3	3	2
	g).	What is the difference between regenerative and dynamic braking?	4	2	2
	h).	State one advantage of chopper drives over conventional methods.	4	2	2
	i).	What is V/f control?	5	2	2
	j).	Define constant torque region.	5	2	2
					5 x 10 = 50 Marks
UNIT-1					
2.	a).	Define Electric drive and explore the components of electric drive system.	1	3	5
	b).	State the advantages of drive system. Illustrate some applications with suitable drive system.	1	3	5
OR					
3.	a).	Selection of suitable drive for various applications.	1	3	5
	b).	Classify different types of Electric braking methods.	1	4	5
UNIT-2					
4.	a).	Demonstrate the four-quadrant operation of drive with neat sketches.	2	2	5
	b).	A Motor drives two loads. One has rotational motion. It is coupled to the motor through a reduction gear with $a=0.1$ and efficiency of 90%. The load has a moment of inertia of 10 kg-m ² and torque of 10 N-m. Other load has translational motion and consists of 1000kg weight to be lifted up at a uniform speed of 1.5 m/sec. Coupling between this load and the motor has an efficiency of 85%. Motor has an inertia of 0.2 kg-m ² and runs at a constant speed of 1420 rpm. Calculate equivalent inertia referred the motor shaft and power developed by the motor.	2	4	5

		OR			
5.	a).	Analyze the steady state stability of Electric drive?	2	4	5
	b).	Explain the load equalization concept.	2	2	5
		UNIT-3			
6.	a).	Demonstrate single phase Fully controlled converter fed separately excited DC Drives for continuous operation with avg. output voltage equation, Speed-Torque relation & voltage and current waveforms.	3	3	5
	b).	A 200V, 875 rpm, 150 A separately excited DC motor has an armature resistance of 0.06Ω . It is fed from single phase fully controlled rectifier with a source voltage of 220V, 50 Hz. Assuming continuous conduction. Calculate , i) Firing angle for rated motor torque & 750 rpm ii) Firing angle for rated motor torque & (-500) rpm iii) Motor speed for $\alpha = 160^\circ$.	3	4	5
		OR			
7.	a).	Explain rectifier fed DC series motor in continues mode operation with equations and speed-torque characteristics.	3	3	5
	b).	A 230V, DC series motor has a resistance of 0.2Ω . At speed of 1800 rpm it takes 50 A. Determine the resistance to be added in series with the motor to limit the speed to 3600 rpm when the current is 12.5 A. Assume magnetizing curve to be a straight line between 0 to 50 A.	3	4	5
		UNIT-4			
8.	a).	Explain the chopper-controlled DC series motor in continues mode operation with necessary equations and speed-torque characteristics.	4	3	5
	b).	Investigate Closed loop speed control scheme of DC Drive for control above and below base speed.	4	3	5
		OR			
9.	a).	Demonstrate the chopper-controlled separately excited DC motor in continues mode operation with equations and speed-torque characteristics.	4	3	5
	b).	230V, 900RPM and 180A separately excited DC motor has an armature resistance of 0.02 ohms. The motor is fed from chopper which provides both motoring and braking operations. The source has a voltage of 230V. Assuming continues conduction. (i) Calculate duty ratio of chopper for motoring operation at rated torque & 380 rpm. (ii) Calculate duty ratio of chopper for braking operation at rated torque & 300 rpm.	4	4	5

		UNIT-5			
10.	a).	Explain with equations and torque-speed characteristics of Induction motor. How can we control speed of an induction motor by AC voltage controller? Why this voltage control method is inefficient?	5	3	5
	b).	What is meant by slip power and how can we extract it? Demonstrate Static scherbius drive with equations and characteristics?	5	3	5
		OR			
11.	a).	Sketch the speed-torque characteristics for an induction motor with Variable Frequency Control.	5	3	5
	b).	Explain about slip power recovery? Discriminate Static Kramer's Drive with circuit diagram.	5	3	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I SemesterMODEL QUESTION PAPER					
POWER SYSTEM PROTECTION					
Electrical & Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State any two essential qualities of a protective system.	1	2	2
	b).	Define “restriking voltage” and mention one effect of high restriking voltage.	1	2	2
	c).	Write any two advantages of SF ₆ circuit breakers over oil circuit breakers.	2	2	2
	d).	Name any two types of circuit breakers used in high-voltage systems.	2	2	2
	e).	Distinguish between primary and back-up protection.	3	2	2
	f).	What is an overcurrent relay? Give one use of it in a power system.	3	2	2
	g).	Name any two types of faults commonly protected in an alternator and briefly mention the protection used.	4	2	2
	h).	What is the main purpose of a surge arrester in a power system?	4	2	2
	i).	Mention any two main features of a numerical relay used in modern substations	5	2	2
	j).	List any two advantages of static relays over electromagnetic relays.	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Describe the essential qualities of a protective relay.	1	3	5
	b).	Distinguish clearly between the recovery voltage and restriking voltage and explain the significance of RRRV in the operation of a circuit breaker by deriving necessary expression.	1	4	5
OR					
3.	a).	Explain the reason for initiation of electric arc during contact separation.	1	3	5
	b).	Explain the phenomenon of current chopping and its effect on circuit interruption.	1	3	5

		UNIT-2			
4.	a).	Describe the operating principle of an air blast circuit breaker with the help of a diagram.	2	3	5
	b).	Describe the construction, operating principle and application of a SF6 circuit breaker	2	3	5
		OR			
5.	a).	Explain the components required for testing of a circuit breaker, and explain the testing procedure.	2	3	5
	b).	Describe the operational phenomena of a vacuum Circuit breaker.	2	3	5
		UNIT-3			
6.	a).	With neat diagram explain the construction and operation of induction type directional over current relay.	3	3	5
	b).	What is universal torque equation? Use this equation and derive the characteristics of impedance relay.	3	3	5
		OR			
7.	a).	Explain the terms (i) Pick-up value, (ii) Reach (iii) Primary protection (iv) Backup protection (v) operating time.	3	3	5
	b).	Explain in detail about the time current characteristics of IDMT relay.	3	3	5
		UNIT-4			
8.	a).	Explain the protection of a generator against (i)stator inter turn fault and(ii) over speeding	4	3	5
	b).	With aid of neat schematic diagram describe the percentage differential protection scheme of a transformer.			
		OR			
9.	a).	Explain in detail about Bucholtz relay with a neat sketch.	4	3	5
	b).	What is the importance of bus-bar protection? What are the requirements of protection of lines?	4	3	5
		UNIT-5			
10.	a).	Explain clearly how a phase comparator is used in the protective relay.	5	3	5
	b).	List the advantages and disadvantages of numerical relays.	5	3	5
		OR			
11.	a).	Describe the basic functional blocks of a numerical relay.	5	3	5
	b).	Explain the operation of rectifier type amplitude comparator in a static relay.	5	3	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EE4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ADVANCED CONTROL SYSTEMS					
Electrical and Electronics Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the need of Compensation.	1	2	2
	b).	What is steady state error? How can steady state error be improved.	1	2	2
	c).	Which compensator improves steady-state accuracy?	2	2	2
	d).	What is Gain Margin?	2	2	2
	e).	Define state feedback control.	3	2	2
	f).	Why is pole placement used in control systems?	3	2	2
	g).	What are the advantages of Digital Computers.	4	2	2
	h).	Write short notes on ZOH.	4	2	2
	i).	Write the bilinear transformation substitution formula.	5	2	2
	j).	State Jurys stability criterion of a discrete-time system in z-domain	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.		A Unity Feedback system has an open-loop transfer function $G(s) = k/s(s+7)$ and is operating with 15% POS in its step response. Using Root locus technique, design a Lag compensator to reduce the SS error for a ramp input by a factor of 20.	1	3	10
OR					
3.		Explain design procedure of PID controllers with examples	1	3	10
UNIT-2					
4.		A UFB system has an OL transfer function with adjustable gain as shown. $G(s) = \frac{k}{s(s+1)}$ Design a Lead compensator using bode approach to meet the specifications of $K_v=10$ and $PM \geq 45^\circ$.	2	3	10
OR					
5.		A unity feed Back system has an open loop Transfer function $G(s) = \frac{k}{s(s+1)(0.2s+1)}$. Design Phase-lag compensation for the system to achieve the following specification. Velocity error constants $K_v=8$, phase margin is 40 degrees. Also compare the crossover frequency of uncompensated and compensated systems.	2	4	10

		UNIT-3			
6.	a).	Explain about pole placement technique with state feedback control.	3	3	5
	b).	Test the controllability and observability for the following SS model $\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t) ; y = [3 \quad 4 \quad 1]$	3	3	5
		OR			
7.	a).	Explain briefly the effect of state feedback on observability.	3	3	5
	b).	Determine the state feedback gain matrix for locating the CL poles of the given system at -2+j4, -2-j4 and -10 $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	3	3	5
		UNIT-4			
8.		Describe the components of a sampled-data control system with a neat block diagram. Explain the roles of ADC, DAC and the digital controller.	4	3	10
		OR			
9.	a).	Explain the operation of a Zero Order Hold (ZOH) circuit and derive its transfer function.	4	3	5
	b).	Derive the pulse transfer function of a sampled-data control system.	4	3	5
		UNIT-5			
10.	a).	The closed loop transfer function for the digital control system is given as $\frac{C(z)}{R(z)} = (z + 0.5)/3(z^2 - z + 0.5)$. Find the steady state errors and error constants due to step input.	5	3	5
	b).	Apply Jury's stability test to determine the stability of a discrete-time system with characteristic equation: $z^3 - 0.5z^2 + 0.3z - 0.1 = 0$.	5	3	5
		OR			
11.	a).	Explain transient response characteristics on z-plane. A digital system has poles at $z = 0.8e^{\pm j45^\circ}$. Determine: i) Damping ratio ii) Natural frequency	5	3	5
	b).	A unity feedback sampled-data system has ZOH and plant $G(s) = K/[s(s+2)]$ with $T = 0.5$ sec and $K = 1$. Derive the closed-loop pulse transfer function $C(z)/R(z)$ and find the characteristic equation.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks